

PD240W Fabrication Document

Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste				Paste Mask	
	F,Silkscreen			Direct Printing	Legend	GBR
	F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
	Prepreg		0,0994mm	FR4_7628	Dielectric	
	Copper	L2 (GND)	0,0152mm (0,43oz)		Plane	GBR
	Core		0,55mm	FR4	Dielectric	
	Copper	L3 (Sig, PWR)	0,0152mm (0,43oz)		Signal	GBR
	Prepreg		0,1088mm	FR4_7628	Dielectric	
	Copper	L4 (Sig, PWR)	0,0152mm (0,43oz)		Signal	GBR
	Core		0,55mm	FR4	Dielectric	
	Copper	L5 (GND)	0,0152mm (0,43oz)		Signal	GBR
	Prepreg		0,0994mm	FR4_7628	Dielectric	
	Copper	L6 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
	B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	B,Silkscreen			Direct Printing	Legend	GBR
	B,Paste				Paste Mask	

Total thickness: 1,5784mm
Notes: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	90	±10 %	L1	0,15	0,2032	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY COMPANY NAME.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

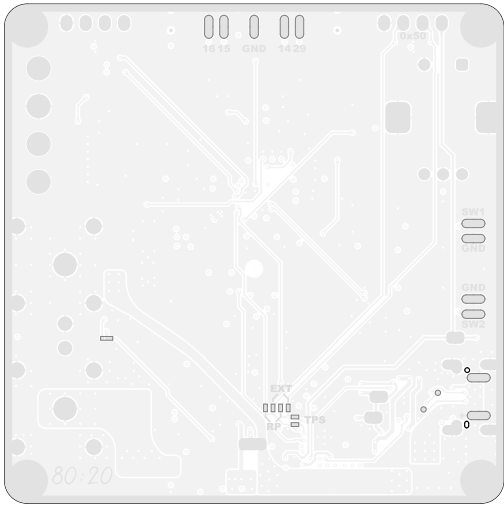
BOARD SIZE	66.000 × 66.000 mm
BOARD THICKNESS	1.578 mm
TRACE WIDTH	0.150 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	0.750 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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PD240W Fabrication Document

Bottom Fabrication (Scale 1:1)



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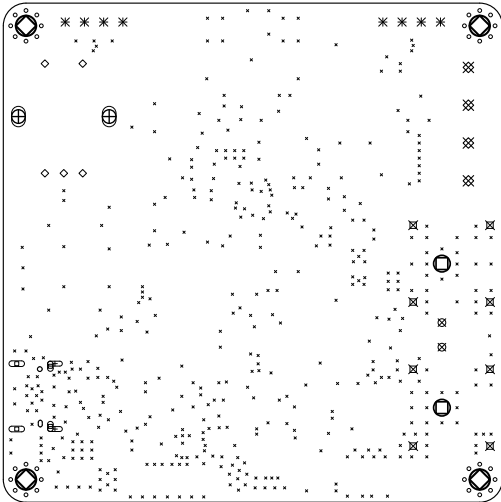
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	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	466	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
○	32	0,50mm (19,69mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
+	2	0,65mm (25,59mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
□	4	0,75mm (29,53mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
◇	5	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊗	2	1,02mm (40,16mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
※	8	1,10mm (43,31mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	8	1,20mm (47,24mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊞	4	1,40mm (55,12mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊕	2	1,80mm (70,87mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	2	2,20mm (86,61mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	4	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
Total 539						

Drill Drawing L1 - L6 (Scale 1:1)



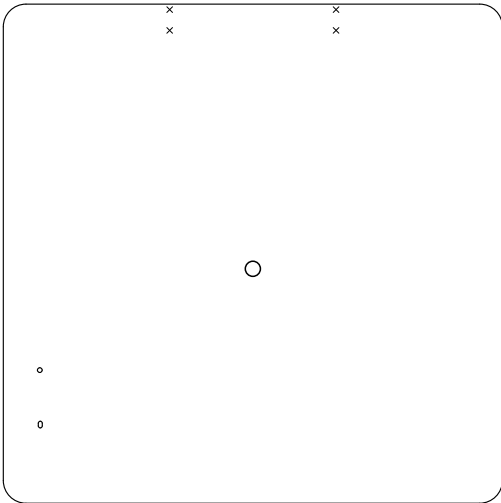
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	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Drill Drawing (L1 - L6)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	0,75mm (29,53mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
O	1	2,00mm (78,74mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
	Total 5					

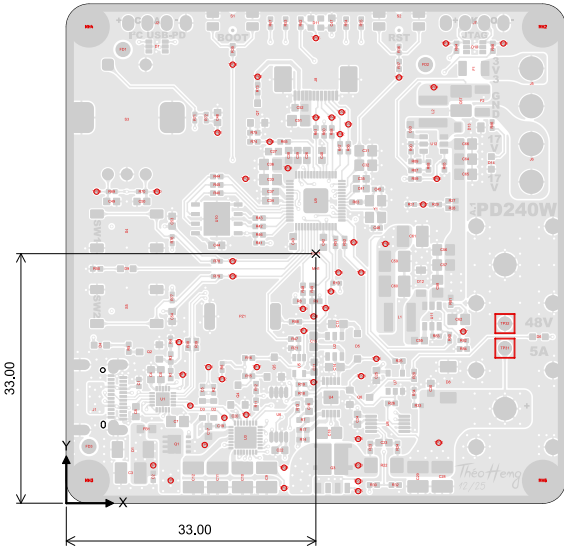
Drill Drawing L1 - L6 (Scale 1:1)



Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Drill Drawing (L1 - L6)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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PD240W Fabrication Document

Top Test Points (Scale 1:1)



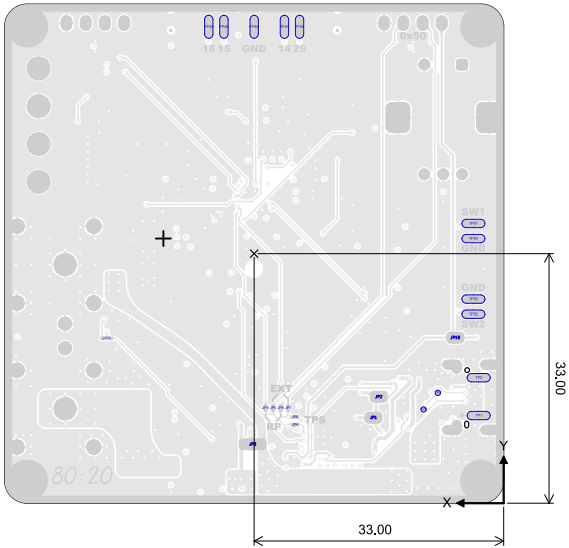
Ref.	Net	X [mm]	Y [mm]	Ref.	Net	X [mm]	Y [mm]
TP3	CC1	16.40	12.90	TP37	RESET	44.00	56.30
TP5	CC2	20.90	11.30	TP38	QSPI_SS	22.00	58.00
TP7	+VBUS_C	11.50	5.00	TP39	I2C0_SDA	31.30	25.70
TP8	+VBUS	13.80	3.00	TP40	I2C0_SCL	32.70	24.70
TP9	+VBUS_LV	18.30	7.70	TP41	LCD_CS	37.30	50.10
TP10	EPR_BLK_G	18.80	18.00	TP42	LCD_SCL	36.20	48.80
TP11	EPR_EN	15.40	17.00	TP43	LCD_SDA	35.10	51.00
TP12	FLT	20.60	16.40	TP44	LCD_RST	33.00	51.00
TP13	+VBUS_LV/2	14.90	21.30	TP45	LCD_RS	36.40	51.70
TP14	POWER_PATH_EN	23.80	11.70	TP55	LED_DATA	33.00	61.50
TP15	PD_GPIO1	36.00	30.50	TP56	BUZZER_PWM_R	34.30	25.80
TP16	USB_PD_IRQ	35.00	28.00	TP57	BUTTON_1	22.00	32.00
TP17	PP_EN1	25.20	18.00	TP58	BUTTON_2	22.00	30.00
TP18	PD_GPIO3	39.00	30.50	TP59	ENCODER_A	4.00	41.20
TP19	PP_EN2	31.40	22.80	TP60	ENCODER_B	12.00	41.20
TP20	+VBUS_SENSED	38.60	5.00	TP61	ENCODER_BUTTON	20.00	49.00
TP21	PP_EN3	32.70	16.00	TP62	LCD_BACKLIGHT	25.30	56.40
TP22	+1V5_LDO	28.80	2.00	TP63	LCD_BACKLIGHT_EN	26.90	47.80
TP23	TS	38.60	2.90	TP64	+17V	48.60	55.00
TP24	+3V3_LDO	28.80	4.70	TP65	+3V3	42.20	34.30
TP25	LTC_GATE	38.60	1.60	TP66	17V_EN	49.00	42.90
TP26	+6V6_LDO	41.00	19.10	TP67	3V3_EN	52.50	22.70
TP27	V_SENSE	47.20	39.50	TP68	3V3_EN_PD	38.60	8.50
TP28	SWITCH_EN	39.00	16.80				
TP29	SW_EN_READ	38.00	34.50				
TP30	+VBUS_OUT	49.25	8.00				
TP31	+VBUS_OUT	58.00	20.50				
TP32	GND	58.00	23.75				
TP33	+1V1	23.70	42.90				
TP34	+3V3	29.30	44.30				
TP35	SWCLK	50.20	60.80				
TP36	SWDIO	57.80	60.80				

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Comments:	Company:		Variant:	Git Hash:
	Company Name		RELEASED	ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Top Test Points (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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Bottom Test Points (Scale 1:1)



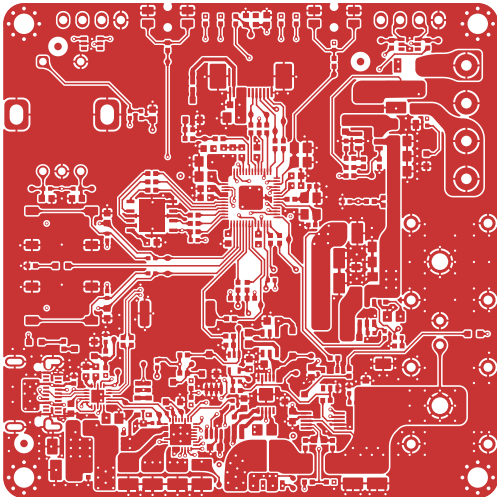
Ref.	Net	X [mm]	Y [mm]
TP1	+VBUS_C	3.30	11.60
TP2	GND	3.30	16.60
TP4	CC1_C	8.70	14.60
TP6	CC2_C	10.60	12.40
TP46	GND	33.00	63.00
TP47	GPIO14	29.00	63.00
TP48	GPIO15	37.00	63.00
TP49	GPIO16	39.00	63.00
TP50	GPIO29	27.00	63.00
TP51	BUTTON_1	4.00	37.00
TP52	BUTTON_2	4.00	25.00
TP53	GND	4.00	27.00
TP54	GND	4.00	35.00

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	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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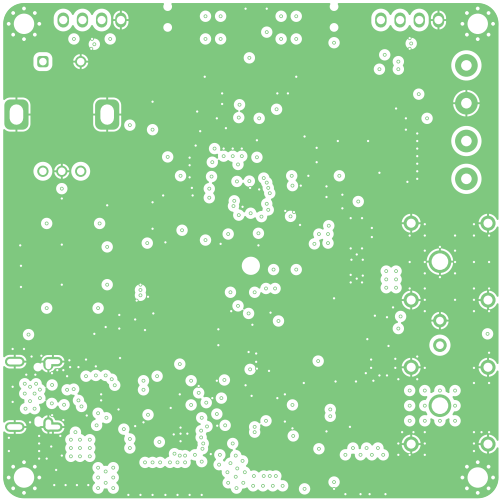
L1 (Sig, PWR) (Scale 1:1)



Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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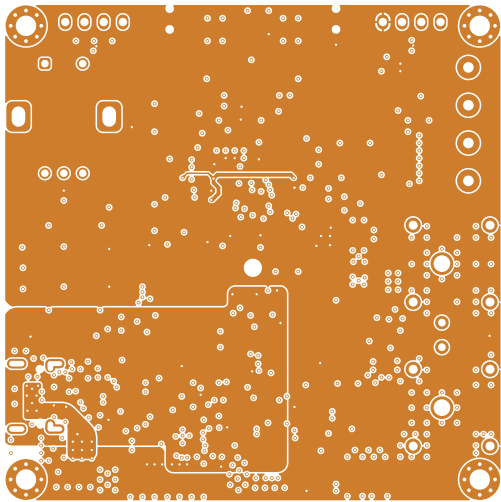
L2 (GND) (Scale 1:1)



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Sheet Title: L2 (GND) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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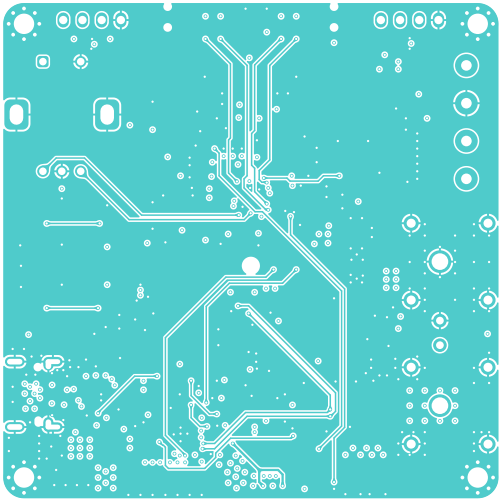
L3 (Sig, PWR) (Scale 1:1)



Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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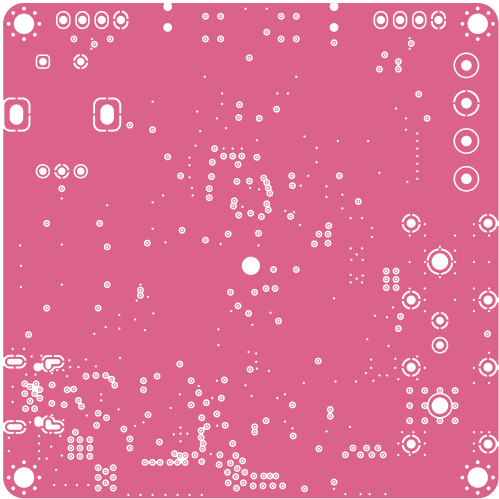
L4 (Sig, PWR) (Scale 1:1)



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Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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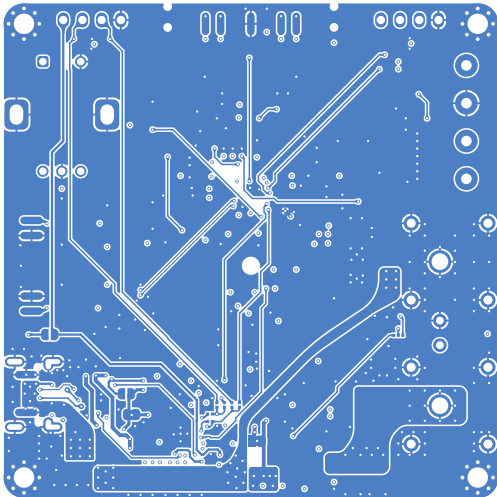
L5 (GND) (Scale 1:1)



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Sheet Title: L5 (GND) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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L6 (Sig, PWR) (Scale 1:1)



Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ac7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: L6 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2024-04-13	Revision: 1.0.0
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